

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-aa5c683f1b9ccb4da.jsonl`  
- SHA-256: `1ea10a3c3c9e49fc31526fb66706567e36e087476a57dcd2804a8257063a42bc`  
- Source modified: `2026-06-16T03:25:38+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:24:30.877Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"TemporalMaxer is a single-stage TAL detector built directly on the ActionFormer anchor-free head (same sequence-labeling decoder where every moment is classified plus onset/offset regression via Focal Loss + DIoU loss), with the ONLY change being that ActionFormer's self-attention temporal-context-modeling block is replaced by a parameter-free MaxPooling block (kernel 3, stride 2) between feature-pyramid levels."

****Source:**** <https://arxiv.org/abs/2303.09055> (primary)

****Supporting quote:**** "Similar to ActionFormer, TemporalMaxer follows a minimalistic design of sequence labeling where every moment is classified, and their corresponding action boundaries are regressed... The sole variation in our study was the substitution of the Transformer block in ActionFormer with the Max Pooling block."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:24:33.693Z)

I'll verify this claim about TemporalMaxer's architecture. Let me fetch the primary source and search for corroborating/contradicting evidence.

3. assistant (2026-06-16T03:24:37.387Z)

Let me fetch the primary arXiv source for the TemporalMaxer paper to verify the specific architectural claims.

4. user (2026-06-16T03:24:43.873Z)

I cannot provide the specific technical details you requested based on the web page content provided.

The abstract states that TemporalMaxer uses "a basic, parameter-free, and local region operating max-pooling block" to replace "long-term temporal context modeling" and "self-attention mechanisms." However, the page content does not include:

1. Whether it uses ActionFormer's anchor-free head
2. Details about classification/regression mechanisms
3. Specific loss functions (Focal Loss, DIOU loss)
4. Exact kernel sizes and strides for max-pooling
5. Technical architectural comparisons with ActionFormer

The abstract only provides this limited detail: "Picking out only the most critical information for adjacent and local clip embeddings, this block results in a more efficient TAL model."

To answer your questions accurately, you would need to access the full PDF paper (available via the "/pdf/2303.09055" link on the page) rather than just the abstract and metadata shown here.

5. user (2026-06-16T03:24:46.938Z)

Web search results for query: "TemporalMaxer max pooling replaces transformer ActionFormer kernel stride architecture"

Links: [{"title": "TemporalMaxer: Maximize Temporal Context with only Max Pooling", "url": "https://arxiv.org/pdf/2303.09055"}, {"title": "TemporalMaxer/README.md at master · TuanTNG/TemporalMaxer", "url": "https://github.com/TuanTNG/TemporalMaxer/blob/master/README.md"}, {"title": "GitHub - f1brahim-tmu/TemporalMaxer: TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization · GitHub", "url": "https://github.com/f1brahim-tmu/TemporalMaxer"}, {"title": "GitHub - zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation: Temporal Action Detection & Weakly Supervised Temporal Action Detection & Temporal Action Proposal Generation · GitHub", "url": "https://github.com/zhenyingfang/Awesome-Temporal-Action-Detection-Temporal-Action-Proposal-Generation"}, {"title": "Efficient Spatial-Temporal Focal Adapter with SSM for Temporal Action Detection This work was supported by JST CRONOS JPMJCS24K4.", "url": "https://arxiv.org/html/2604.09164"}, {"title": "Benchmarking Data Efficiency and Computational Efficiency of Temporal Action Localization Models", "url": "https://arxiv.org/pdf/2308.13082"}, {"title": "A Two-stage Transformer Framework for Temporal Localization of Distracted Driver Behaviors", "url": "https://arxiv.org/pdf/2603.21048"}, {"title": "[2303.09055] TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization", "url": "https://arxiv.org/abs/2303.09055"}]

Based on the search results, here's what I found about TemporalMaxer:

Overview

TemporalMaxer is a method that minimizes long-term temporal context modeling while maximizing information from extracted video clip features with a basic, parameter-free, and local region operating max-pooling block.

Relationship to ActionFormer

Max Pooling achieves the highest results at every tIoU threshold and significantly outperforms the strong and robust baseline, ActionFormer. More specifically, TemporalMaxer improves the robust baseline ActionFormer by 2.4% mAP at tIoU=0.7, and 1.3% mAP on average.

Architecture Approach

Instead of employing a computationally-heavy backbone such as a Transformer, the model uses a basic, parameter-free max pooling block on top of a pre-trained 3D CNN. This is distinct from ActionFormer, which is built on the transformer architecture.

Efficiency Gains

TemporalMaxer is effective at modeling temporal contexts, outperforming the robust baseline ActionFormer with 2.8x fewer GMACs and 3x faster inference speed. Additionally, TemporalMaxer only takes 2.5 ms, which is 8.0x faster than ActionFormer backbone at 20.1 ms.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:24:50.978Z)

The abstract confirms the high-level claim but I need the full PDF to verify the specific details: the anchor-free head, Focal Loss + DIoU loss, and the exact kernel-3/stride-2 MaxPooling parameters. Let me fetch the full PDF and the GitHub README.

7. user (2026-06-16T03:24:58.376Z)

I cannot extract the requested technical architecture details about TemporalMaxer from the provided content. The document appears to be a PDF containing compressed binary data and embedded images that cannot be reliably parsed as plain text. The content shown is primarily FlateDecode-compressed streams and PDF object references rather than readable technical documentation.

To obtain accurate verbatim quotes about TemporalMaxer's architecture, loss functions, pooling specifications, and implementation details, you would need to:

1. Provide the document in a readable text or clean PDF format
2. Share the actual research paper or technical report directly
3. Access the original source material through academic databases

I cannot responsibly extract or invent technical specifications for a system, as doing so could spread misinformation about its actual design and performance characteristics.

[Binary content (application/pdf, 1.1MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579965172-50v5x2.pdf]

8. user (2026-06-16T03:25:00.545Z)

TemporalMaxer vs ActionFormer and Architecture Details

Based on the README provided, here's what is addressed:

****ActionFormer Comparison:****

The document states that TemporalMaxer "outperforms the robust baseline, ActionFormer, with 2.8x fewer GMACs and 3x faster inference speed" and notes the backbone is "8.0x faster than ActionFormer backbone."

****Max Pooling Block Specifications:****

The README does not provide kernel size or stride details for the max pooling operation.

****Loss Functions and Head Design:****

The README makes no mention of Focal Loss, DIoU loss, anchor-free approaches, or sequence labeling heads. It only describes the method as containing a "parameter-free, and local region

operating max-pooling block" but lacks the granular architectural specifications you've asked about.

The document emphasizes simplicity and efficiency gains over ActionFormer but omits the technical implementation details regarding loss functions and detection head design that you're inquiring about.

9. assistant (2026-06-16T03:25:02.744Z)

The PDF was saved locally. Let me read it directly to extract the verbatim technical details, and also check the HTML version on arXiv.

10. user (2026-06-16T03:25:04.167Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

11. user (2026-06-16T03:25:06.262Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

12. assistant (2026-06-16T03:25:08.040Z)

Let me extract text from the PDF using a Python-based approach.

13. user (2026-06-16T03:25:13.880Z)

```
=== ActionFormer ===
deo Transformer Head Pretrained 3D CNN Input video TemporalMaxer Head Pretrained 3D CNN Input
video Convolution Head Pretrained 3D CNN Common Architecture AFSD ActionFormer Ours MaxPooling
Clip embeddingsInput video Graph Head Pretrained 3D CNN G-TAD Figure 1. Common architecture in
Temporal Action Localization (TAL) and different
=== sequence labeling ===
that utilizes a state-of-the-art Action-Former [60] as the baseline for comparison. Similar to
ActionFormer, TemporalMaxer follows a minimalistic design of sequence labeling where every
moment is classified, and their corresponding action boundaries are regressed. The main
difference is that we avoid exhausting attention between c
=== Max Pooling ===
TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization
Tuan N. Tang, Kwonyoung Kim, Kwanghoon Sohn* School of Electrical and Electronic Engineering
Yonsei University {tuantng, kyk12
=== max pooling ===
TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization
Tuan N. Tang, Kwonyoung Kim, Kwanghoon Sohn* School of Electrical and Electronic Engineering
Yonsei University {tuantng, kyk12
=== Focal ===
e multi-scale feature pyramidZ. Learning Objective . The model predicts  $\hat{t} = (o, e, t, c, t)$ 
for every moment of the inputX. Following the baseline [60], the Focal Loss [31] and DIOU loss
[64] are employed to supervise classification and regression outputs respectively. The overall
loss function is defined as:  $L_{total} = \sum t$ 
=== DIOU ===
e pyramidZ. Learning Objective . The model predicts  $\hat{t} = (o, e, t, c, t)$  for every moment of
the inputX. Following the baseline [60], the Focal Loss [31] and DIOU loss [64] are employed to
supervise classification and regression outputs respectively. The overall loss function is defined
as:  $L_{total} = \sum t (L_{cls} + 1 \cdot L_{reg})/T$ 
=== kernel ===
The sole variation in our study was the substitution of the Transformer block in ActionFormer
with the Max Pooling block. All experiments are conducted with a kernel size of 3 for all TCM
blocks. The subsequent ablation will thorough analysis of the effects of varying kernel sizes.
During training, the input feature length
=== stride ===
ly, the multi-scale temporal feature pyramidZ is encoded by TemporalMaxer:  $Z_l = \text{TemporalMaxer}(Z$ 
```

l?1). (4) Here: TemporalMaxer is Max Pooling and employed with stride 2, Z l? R T 2l?1 ×D, 2 ≤ l ≤ L. It is worth noting that ActionFormer [60] employs Transformer [51] as a TCM block where each clip feature at the moment t r

=== anchor ===

challenging but essential task within the field of video comprehension. Various approaches have been proposed to address this task, such as action proposals [29], anchor windows [36], or dense prediction [28]. A widely accepted notion for improving the performance of TAL models is to integrate a component capable of capturing

=== classify ===

calization (TAL). Two-Stage and Single-Stage methods are used in TAL to detect actions in videos. Two-Stage methods first generate possible action proposals and classify them into actions. The proposals are generated through anchor windows [13, 6, 20], detecting action boundaries [30, 17, 62], graph representation [4, 54], o

=== regress ===

ActionFormer, TemporalMaxer follows a minimalistic design of sequence labeling where every moment is classified, and their corresponding action boundaries are regressed. The main difference is that we avoid exhausting attention between clips from long-term timeframes which can unintentionally attend minute information am

=== sole ===

much faster inference speed. Our finding suggests that the pre-trained feature extractor already possess great potential and with those feature, short-term TCM solely can benefit the performance for this task. Extensive experiments prove the superiority and effectiveness of the proposed method, showing state-of-the-art

=== only ===

TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization
Tuan N. Tang, Kwonyoung Kim, Kwanghoon Sohn* School of Electrical and Electronic Engineering
Yonsei University {tu

=== feature pyramid ===

emporalMaxer Figure 3. Overview of TemporalMaxer. The proposed method utilizes Max Pooling as a Temporal Context Modeling block applied between temporal feature pyramid levels to maximize informative features of high similarity clip embedding. Specifically, it first extracts features of every clip using pre-trained 3D CNN. After

14. assistant (2026-06-16T03:25:27.605Z)

I now have all the verbatim evidence needed from the primary source PDF. Let me verify each component of the claim against the extracted text.

The claim has five sub-assertions. Checking each against the verbatim PDF text:

1. *****built directly on the ActionFormer anchor-free head (same sequence-labeling decoder where every moment is classified plus onset/offset regression)***** ? CONFIRMED verbatim: "Similar to ActionFormer, TemporalMaxer follows a minimalistic design of sequence labeling where every moment is classified, and their corresponding action boundaries are regressed." The model predicts $\hat{t} = (os_t, oe_t, c_t)$ for every moment (onset/offset/class) ? this is the anchor-free dense prediction head.
2. *****via Focal Loss + DIOU loss***** ? CONFIRMED verbatim: "Following the baseline [60], the Focal Loss [31] and DIOU loss [64] are employed to supervise classification and regression outputs respectively."
3. *****the ONLY change being that ActionFormer's self-attention temporal-context-modeling block is replaced by a parameter-free MaxPooling block***** ? CONFIRMED verbatim: "The sole variation in our study was the substitution of the Transformer block in ActionFormer with the Max Pooling block." Also: "The main difference is that we avoid exhausting attention between clips" and the architecture figure shows the TCM (Temporal Context Modeling) block swap.
4. *****kernel 3, stride 2***** ? CONFIRMED verbatim: "All experiments are conducted with a kernel size of 3 for all TCM blocks" and "TemporalMaxer is Max Pooling and employed with stride 2".
5. *****between feature-pyramid levels***** ? CONFIRMED verbatim: "The proposed method utilizes Max Pooling as a Temporal Context Modeling block applied between temporal feature pyramid levels" and equation: $Z_l = \text{TemporalMaxer}(Z_{l+1})$.

Every single sub-claim is directly supported by verbatim text from the primary source (arXiv

2303.09055). The supporting quote in the claim is accurate. The source is the primary paper itself (peer-reviewed CVPR Workshop 2023 / arXiv). No contradicting evidence exists; the GitHub repo and search results corroborate the high-level swap.

One minor nuance to flag adversarially: the claim says "parameter-free MaxPooling block" ? confirmed by abstract ("parameter-free... max-pooling block"). The claim says "anchor-free head" ? the paper says "sequence labeling where every moment is classified and boundaries regressed," which is precisely the anchor-free dense-prediction formulation ActionFormer popularized; this is an accurate characterization, not an overreach. All technical specifics (kernel 3, stride 2, Focal+DIoU, sole change) are verbatim-confirmed.

The claim is fully supported by the primary source, current (2023, still SOTA-class), and the source quality (primary arXiv paper) matches the claim strength.

15. user (2026-06-16T03:25:35.229Z)

Structured output provided successfully

16. assistant (2026-06-16T03:25:38.850Z)

Claim verified as NOT refuted with high confidence. Every sub-assertion in the claim is confirmed verbatim from the primary source (arXiv 2303.09055, read directly from the downloaded PDF): the ActionFormer sequence-labeling/anchor-free head, Focal Loss + DIoU loss, the "sole variation" being the Transformer?MaxPooling swap, kernel size 3, stride 2, and application between feature-pyramid levels. No contradicting evidence found; corroborated by the official GitHub repo and search results.